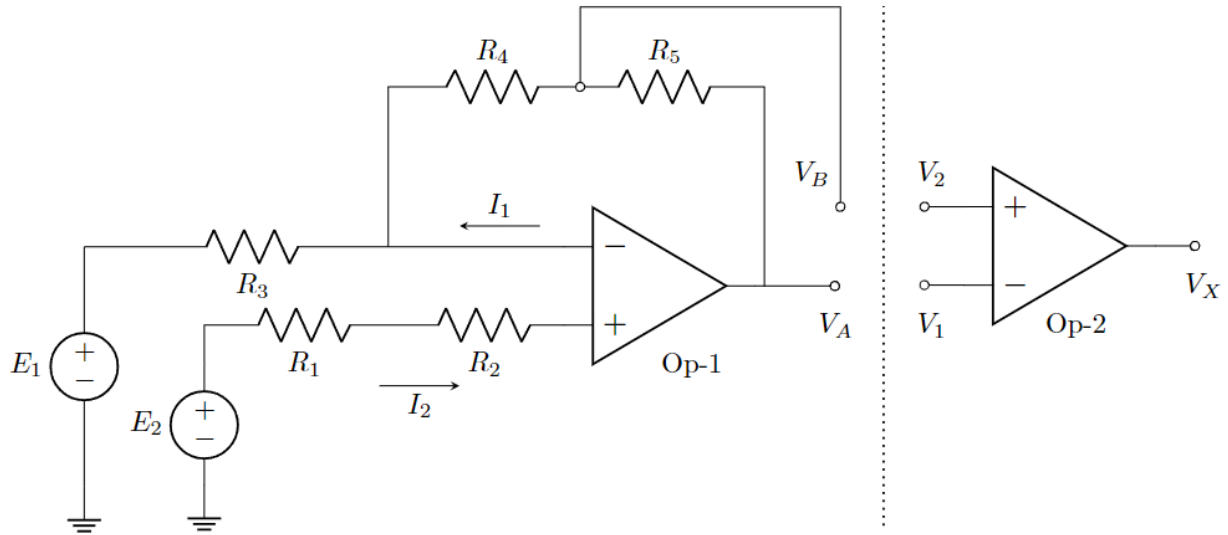


BRAC UNIVERSITY
Department of Computer Science and Engineering
 Assignment 01, Section: 06, Summer 2024
 CSE 251: Electronic Devices and Circuits

1.



The ‘ideal’ operational amplifiers (Op-Amp) have been connected to saturation voltages $V_{sat}^+ = +8V$ and $V_{sat}^- = -8V$. The resistor values are given as,

$$R_1 = R_2 = 1k\Omega, \text{ and } 4R_4 = 10R_5 = 20k\Omega$$

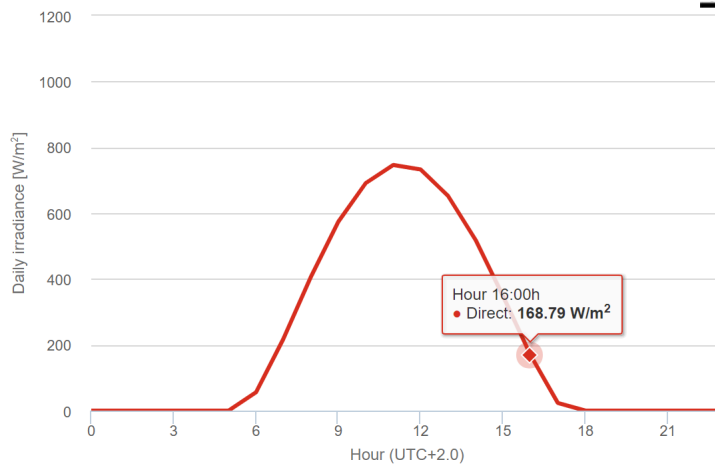
(a) What are the current values of I_1 and I_2 ? [1]

(b) If $E_1 = 5V$, $E_2 = 0V$, and $R_3 = 10k\Omega$, determine V_A and V_B . [2+2]

(c) For the setup of question (b), find the value of V_X when we take $V_1 = V_A$ and $V_2 = V_B$. [2]

(d) For $E_1 = 0V$ and $E_2 = 2.2V$ we measure $V_A = 5.13V$. Calculate what value of R_3 will make this possible. [3]

2.



The above figure shows the variation of daily average solar irradiance (in units of W/m^2) throughout the day in Egypt. In the legend, we can see that at 16:00h, the direct solar irradiance is about 168.79 W/m^2 .

A photodetector module that produces a voltage V_s across its terminal as per the following relation, is used to convert solar irradiance G (in units of kW/m^2) to usable voltage.

$$V_s(G) = 5G^2 + 8G$$

You are asked to **design** a circuit using an Op-Amp that will be connected to a *street-lamp (or an LED)* and that will **turn the lamp (LED) ON after 17:00h**. The lamp (LED) turns **ON** at **5 V**, and remains **OFF** at **0 V**.

- Determine whether the Op-Amp comparator circuit needed will be in inverting, or non-inverting configuration. [2]
- Determine the **threshold (reference) voltage** with which the input voltage is compared. [3]
- Draw the completed Op-Amp comparator circuit. Clearly indicate input, output, terminals, saturation and reference voltages. [5]